

What Can a VLA Learn from Synthetic Depth? Distilling Geometry from Human Demonstrations

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Abstract

Vision-language-action (VLA) models inherit their visual pathway from web-scale pretrained RGB encoders. Strong depth and 3D foundation models exist, but they are optimized for geometric reconstruction rather than action semantics, and no depth encoder is aligned to manipulation the way language-image pretraining aligns RGB. Geometry is therefore usually added to a policy either by training a 3D encoder from scratch on scarce robot data or by assuming a calibrated depth sensor at deployment, and neither route is open to a practitioner holding a frozen, already-working RGB backbone. Even with a depth image in hand at test time, there is no trained module that turns it into a feature the policy can read. We present D-LAPA, which *builds* the missing depth encoder by distillation instead of assuming one. A depth latent-action teacher is trained once to model geometric change between frames; a lightweight single-timestep encoder is then distilled from its pseudo-labels while conditioning on frozen LAPA RGB features and one depth map. Both stages learn exclusively from unlabeled human demonstration video (Something-Something V2): no robot trajectories, no teleoperation, no action labels, and no depth sensor at any point in the pipeline. The resulting 1024-dimensional feature is concatenated with the RGB feature during downstream action fine-tuning. On the LIBERO benchmark, D-LAPA improves over the LAPA baseline in all six suite-by-variant cells, raising the three-suite average from 58.0% to 63.7% (token supervision) and 62.7% (feature distillation). Feature analysis further indicates that depth produces a more locally coherent representation of movement magnitude, suggesting that synthetic geometry can provide complementary action-relevant structure beyond RGB pretraining.

Introduction

Learning robot manipulation policies from large-scale demonstration data is bottlenecked by the cost of collecting action-labeled trajectories (Kim et al. 2024; Bjorck et al. 2025). Latent action pretraining (LAPA) (Ye et al. 2025) offers an appealing alternative: it treats the visual change between two video frames as an implicit action, quantizes that transition into a short sequence of discrete latent codes,

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and later re-grounds these codes into real robot actions with a modest amount of downstream fine-tuning. Because the pretraining signal is derived entirely from unlabeled RGB video, LAPA can exploit web-scale human video such as Something-Something V2 (Goyal et al. 2017). While the RGB feature is a single-frame, language-conditioned representation, the depth feature may describe geometric changes, which has been shown in Fig. 1.

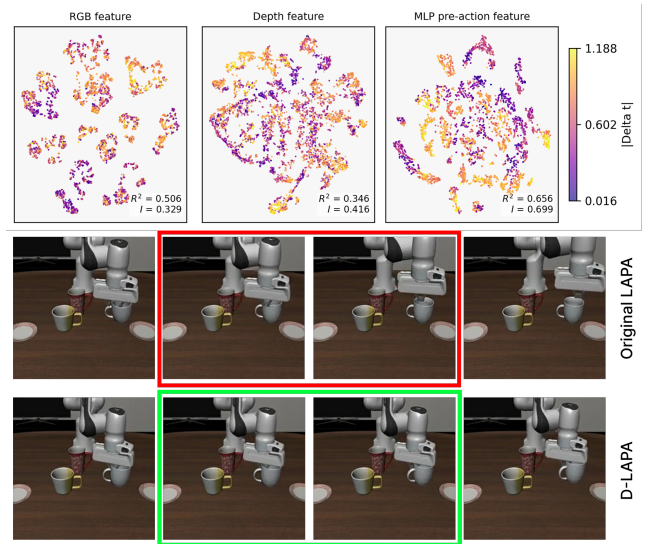


Figure 1: **Top**: the three representations available to a D-LAPA policy on LIBERO-Goal, embedded and coloured by the movement magnitude $|\Delta t|$ of the corresponding action—the frozen pretrained LAPA RGB feature, the depth feature emitted by our Stage-2.5 encoder (Model 4), and the pre-action feature at the last layer of the action head that consumes both. R^2 is an in-sample ridge probe ($\lambda = 5000$) predicting $|\Delta t|$; I is Moran’s I of $|\Delta t|$ over the embedding.

Why not simply hand the policy a depth image? The obstacle is not depth but the *encoder*. A VLA’s visual pathway is a pretrained artifact: it works because someone already trained it on far more data than any robotics lab holds. Depth has no equivalent. Excellent depth and 3D foundation models are available (Yang et al. 2024; Wang et al. 2025), but they are trained for geometric reconstruction, without language or ac-

tion supervision, so their features are not aligned to manipulation semantics in the way contrastive image-text pretraining aligns RGB. The systems that do exploit geometry therefore either train a geometric pathway from scratch on the limited robot data available (Yuan et al. 2025; Li et al. 2025c), or feed sensor depth into a policy learned end-to-end (Ze et al. 2024; Li et al. 2025a). For a frozen RGB backbone with no robot RGB-D corpus, both routes are closed: even with a perfect depth map at test time, there is no trained module that converts it into something the policy can consume.

This paper asks a simple question: *can a VLA learn things from synthetic depth information about human demonstrations, and apply that knowledge to process a depth image into useful features for downstream manipulation tasks?* We do not assume the answer. Depth is an intuitively appealing signal for a spatial task, but appealing signals do not automatically become useful features, and it is entirely possible that a frozen RGB latent-action space already implies whatever a policy needs to know about geometry, leaving nothing for a learned depth-processing module to add. Testing the question therefore requires two things: a way to let a model learn, from synthetic depth observed during pretraining, how to turn a depth image into a feature; and a way to isolate whatever that learned feature contributes from other changes to the model, so that any downstream improvement can be attributed to the feature itself rather than to added capacity, extra data, or a different training budget.

Our contributions are fourfold. (1) We test whether a VLA can learn from synthetic depth information about human demonstrations and apply that knowledge to process a depth image into useful features for downstream manipulation, isolating that effect from changes in model capacity or training data. (2) We give a framework that use a depth encoder built by teacher-student distillation entirely from human video, without robot data, action labels, or a depth sensor at any stage, requiring no changes to the pretrained RGB stack. (3) We use movement magnitude as a probe target to perform a feature-level analysis of the fine-tuned policies, examining how the depth-augmented and RGB-only representations differ internally.

Related Work

Latent action pretraining from human video. A growing line of work learns action abstractions directly from video by modeling frame-to-frame transitions, then transfers them to control (Baker et al. 2022; Bruce et al. 2024; Schmidt and Jiang 2024). LAPA (Ye et al. 2025) is the instance we build on: a quantization model (van den Oord, Vinyals, and Kavukcuoglu 2017) compresses the transition between two RGB frames into discrete codes, and a vision-language-action policy is later fine-tuned to emit real actions. Related designs vary the latent parameterization, including motion tokens (Chen et al. 2025b), task-centric latents (Bu et al. 2025), image-goal units (Chen et al. 2024), and continuous rather than quantized codes (Liang et al. 2025; Yang et al. 2025). Other approaches improve how latents are consumed downstream (Chen et al. 2025a; Li et al. 2025d). Analyses of what these latents actually capture find them sensitive to distractors and to the choice of objective (Zhang et al. 2025a;

Nikulin et al. 2025; Polubarov et al. 2025). What unites this literature is its modality: with the recent exception of Govind et al. (2026), discussed below, these methods learn from RGB alone and never observe scene geometry. D-LAPA retains this paradigm and its data assumption. Our depth pathway is likewise learned from unlabeled human video and augments the representation from the outside.

Depth and 3D in manipulation policies. Geometric input is standard in policies trained from scratch (Ze et al. 2024; Zhen et al. 2024), and a rapidly growing body of work adds it to vision-language-action models. The approaches differ mainly in where geometry enters and what it costs at deployment. Pang et al. (2024) inject depth into a *pretrained* RGB policy and explicitly frame it as cross-modal distillation, training with RGB-D but deploying on RGB alone by hallucinating depth features. Li et al. (2025c) supervise a VLA with quantized depth tokens derived from a monocular estimator; Yuan et al. (2025) fuse a purpose-trained depth-prediction transformer into the policy; and Li et al. (2025b) and Zhao et al. (2025) align intermediate VLA features to a 3D foundation model (Wang et al. 2025) without explicit depth input. Point-cloud methods (Li et al. 2025a; Fan et al. 2026) take the opposite trade: Li et al. (2025a) notably freeze the pretrained action expert and inject 3D through a lightweight module, but require a depth sensor at execution. World-model VLAs forecast depth among several future cues (Zhang et al. 2025b). Two observations follow. First, avoiding a deployment depth sensor is now common rather than distinctive. Second, every one of these systems learns its geometric pathway from *robot* data, specifically teleoperated trajectories with action labels. This is precisely the resource that latent action pretraining exists to avoid.

Cross-modal distillation of geometry. Supervising a student in one modality with a teacher trained in another is well established (Hinton, Vinyals, and Dean 2015), and the works above show it is already the dominant mechanism for adding geometry to policies (Pang et al. 2024; Li et al. 2025b; Zhao et al. 2025). D-LAPA differs in what supplies the teacher. Rather than distilling from a geometry model (Wang et al. 2025) or from depth paired with robot trajectories, we distill from a depth *latent-action* teacher: a model trained with LAPA’s quantization objective on depth transitions, so its targets describe geometric *change* in an action-centric latent space rather than static structure. The distillation also serves a second purpose beyond transfer. The teacher consumes a frame pair and could never run in a control loop; the student is restricted to a single frame and therefore can.

Positioning. The work closest to ours is Govind et al. (2026), which is concurrent and likewise brings depth into latent action learning from video. It learns a shared RGB-D latent action space and uses it to pretrain a VLA, but depth is a *training-time* signal only: at inference the policy is RGB-only, and the geometric information survives solely as it was absorbed into the RGB latent space during pretraining. D-LAPA instead keeps the depth pathway active at execution, with monocular depth recomputed from the policy’s own observation at every control step. Table 1 places the closest

Table 1: Systems that are relevant to D-LAPA.

Method	Geometry from	Depth at test	Backbone
DP3, 3D-VLA	robot 3D	sensor	trained
PointVLA	robot 3D	sensor	frozen
Depth Helps	robot RGB-D	none	tuned
QDepth-VLA	robot video	none	trained
Spatial Forcing	robot video	none	trained
UniLACT	robot RGB-D	none	trained
D-LAPA (ours)	human video	monocular	tuned

systems along the three axes that matter for our setting: what data the geometric pathway is learned from, what it costs at deployment, and whether the pretrained backbone survives untouched. What we have not found elsewhere is their conjunction: a geometric pathway acquired *only* from unlabeled human video, with no depth sensor at any stage, distilled into a single-frame student that remains queryable online, and attached to a pretrained VLA in finetuning stage.

Method

Problem setup. Each pretraining sample is a temporal transition $(I_t, I_{t+\Delta}, D_t, D_{t+\Delta})$ of an RGB frame pair drawn from human demonstration video and its aligned depth pair, where the depth frames are estimated monocularly from the RGB frames by an off-the-shelf estimator. No stage of the pipeline consumes sensed depth, robot trajectories, or action labels. The frozen LAPA model provides, for the RGB pair, a continuous transition feature $z_t^{\text{rgb-feat}}$ and discrete latent indices $z_t^{\text{rgb-idx}}$. Our goal is to produce an additional depth-aware feature that can be concatenated with $z_t^{\text{rgb-feat}}$ for downstream policy learning, subject to two constraints: the LAPA stack stays frozen, and the depth pathway stays frozen during policy fine-tuning.

The pipeline has three stages. **Stage 1** trains a depth teacher and freezes it. **Stage 2.5** distills a lightweight depth-aware encoder from the teacher’s pseudo-labels. **Stage 3** fine-tunes the policy under the standard LAPA protocol with the depth encoder frozen.

Stage 1: depth latent-action teacher. The teacher mirrors LAPA’s latent action quantization (LAQ) objective (Ye et al. 2025; van den Oord, Vinyals, and Kavukcuoglu 2017), but on depth. Its encoder maps the depth pair $(D_t, D_{t+\Delta})$ to a continuous transition feature $z_t^{\text{depth-feat}}$, which a vector quantizer discretizes into a length-4 code sequence $z_t^{\text{depth-idx}}$ over a codebook of size $K = 8$. A decoder reconstructs the future frame $D_{t+\Delta}$ from the latent and the current frame D_t , forcing the latent to capture the geometric change between frames. After training, the teacher is frozen; its only role is to emit the two pseudo-label types, $z_t^{\text{depth-feat}}$ and $z_t^{\text{depth-idx}}$, for Stage 2.5.

Stage 2.5: distilling a depth-aware encoder. The Stage-2.5 encoder is deliberately small and, more importantly, deliberately *causal*: it observes single-timestep inputs, never the future frame, and outputs a 1024-dimensional feature. This

Table 2: Stage-2.5 variants. All are supervised by pseudo-labels from the same frozen Stage-1 depth teacher and trained on identical data. CE denotes cross-entropy over depth tokens; MSE+cos denotes continuous feature distillation.

Variant	Input	Target	Loss
Model 2	$D_t + z_t^{\text{rgb-feat}}$	$z_t^{\text{depth-idx}}$	CE
Model 4	$D_t + z_t^{\text{rgb-feat}}$	$z_t^{\text{depth-feat}}$	MSE+cos

asymmetry is the crux of the recipe. The teacher’s supervision is rich precisely because it sees a transition $(D_t, D_{t+\Delta})$, but a model that requires the next frame can never run in a control loop. Distillation transfers what the teacher knows about geometric change into a student that reads only what is available at execution time. The product is therefore not merely a compressed teacher but a new kind of module: a depth encoder whose output is aligned with an action-centric latent space and which is queryable online, one frame at a time. The three retained variants differ along the two design axes introduced above and are summarized in Table 2.

Model 2: token prediction from depth and RGB. The encoder consumes the current depth map and the frozen RGB feature, and a linear head predicts logits over the depth codebook:

$$h_t^D = f_\phi^{(2)}(D_t, z_t^{\text{rgb-feat}}), \quad \hat{\ell}_t^D = g_\phi^{(2)}(h_t^D), \quad (1)$$

with $\hat{\ell}_t^D \in \mathbb{R}^{4 \times K}$, trained by cross-entropy against the teacher tokens:

$$\mathcal{L}_{M2} = \text{CE}(\hat{\ell}_t^D, z_t^{\text{depth-idx}}). \quad (2)$$

The pre-head feature h_t^D serves as the depth representation passed to the policy. Note that the supervision here is indirect: h_t^D is shaped only by what is needed to classify the four depth tokens.

Model 4: continuous feature distillation from depth and RGB. The encoder takes the same inputs but regresses the teacher’s continuous feature directly:

$$\hat{z}_t^D = f_\phi^{(4)}(D_t, z_t^{\text{rgb-feat}}), \quad (3)$$

$$\mathcal{L}_{M4} = \|\hat{z}_t^D - z_t^{\text{depth-feat}}\|_2^2 + \lambda_{\cos} (1 - \cos(\hat{z}_t^D, z_t^{\text{depth-feat}})), \quad (4)$$

with $\lambda_{\cos} = 0.1$. The MSE term anchors feature magnitude; the cosine term aligns direction, which matters when the feature is later consumed by a policy through concatenation.

Stage 3: downstream action fine-tuning. We follow the LAPA action fine-tuning protocol (Ye et al. 2025). The latent action head is replaced by a newly initialized robot action head predicting a seven-dimensional action (end-effector translation, end-effector rotation, and a gripper command); the LAPA vision encoder and VQGAN (Esser, Rombach, and Ommer 2021) stay frozen throughout. At each step, the only inputs are the RGB observation I_t and the language instruction ℓ . The frozen visual branch processes I_t and feeds

two copies of the LAPA LLM backbone. One copy is kept exactly at its pretrained weights (based on SSv2) and produces $z_t^{\text{rgb-feat}} = f_{\text{rgb}}^{\text{pre}}(I_t, \ell)$; the other is fine-tuned over the course of Stage 3 and produces $z_t^{\text{rgb}} = f_{\text{rgb}}^{\text{ft}}(I_t, \ell)$. We keep the pretrained copy around for a specific reason: the Stage-2.5 depth encoder was trained to condition on $z_t^{\text{rgb-feat}}$ as it existed *before* any Stage-3 fine-tuning, so feeding it the fine-tuned feature instead would shift its input away from the distribution it was distilled on. In parallel, a frozen, off-the-shelf monocular depth estimator turns the same observation I_t into a synthetic depth map D_t , and the frozen Stage-2.5 encoder combines it with the frozen RGB feature to produce $\hat{z}_t^D = f_{\text{depth}}(D_t, z_t^{\text{rgb-feat}})$. The policy input concatenates the depth feature with the *fine-tuned* RGB feature, not the frozen one:

$$z_t^{\text{policy}} = [z_t^{\text{rgb}}; \hat{z}_t^D]. \quad (5)$$

This concatenation is what the newly initialized action head consumes to predict $\mathbf{a}_t$. Only the fine-tuned LLM copy and the action head are updated during Stage 3; the visual branch, the pretrained LLM copy, the depth estimator, and the Stage-2.5 encoder are all frozen. Because every variant adds a feature of identical width through an identical frozen interface, the comparison across variants, including the comparison against the baseline, is a comparison of representations rather than model capacity or optimization budget. Figure 2 summarizes the resulting loop. The RGB-only baseline follows the same protocol without the depth branch, and needs only the fine-tuned LLM copy, since there is no depth encoder left to condition on the frozen one.

Experiments

Pretraining data. Stage-1 and Stage-2.5 models are trained on the first 30k videos of Something-Something V2 (SSv2) (Goyal et al. 2017), a corpus of unlabeled human demonstration video, using temporal pairs with offset $\Delta = 30$. SSv2 provides no depth channel; depth frames are estimated monocularly from the RGB frames (Yang et al. 2024) with the same estimator used at deployment, so the training and inference distributions of the depth input are matched by construction and the pipeline never requires a depth sensor. RGB pairs are denoted $(I_t, I_{t+\Delta})$ and aligned depth pairs $(D_t, D_{t+\Delta})$. Depth maps are resized to 256×256 and normalized by their storage range (16-bit maps are divided by 65535). Frozen LAPA RGB features are pre-extracted once and loaded from a feature manifest during Stage-2.5 training.

Finetuning We fine-tune each downstream policy for 20,000 optimization steps using a global batch size of 128. Optimization is performed with AdamW, a constant learning rate of 2×10^{-5} , zero weight decay, and no learning-rate warmup. All models are trained using bfloat16 precision. We independently fine-tune a separate policy for each LIBERO suite using the corresponding demonstrations and suite-specific action-discretization statistics.

Evaluation protocol. We evaluate downstream manipulation on the LIBERO benchmark’s (Liu et al. 2023) Spatial,

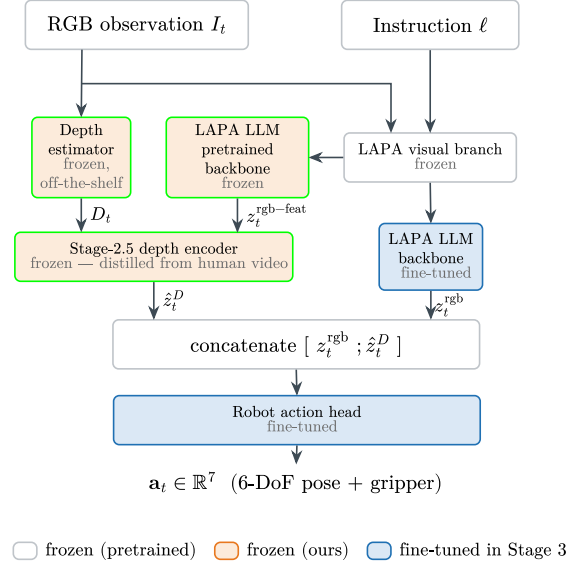


Figure 2: Stage-3 inference and fine-tuning. A frozen depth estimator turns the policy’s own RGB observation into a synthetic depth map; the frozen Stage-2.5 encoder converts it, together with the RGB feature from a *frozen, pretrained* copy of the LAPA LLM backbone, into a depth-aware feature $\hat{z}_t^D$. This feature is concatenated with the RGB feature from a *separate, fine-tuned* copy of the same backbone and passed to the newly initialized action head. Keeping a frozen backbone copy for the depth encoder preserves the input distribution it was distilled on; only the fine-tuned backbone copy and the action head are updated in Stage 3. No depth sensor and no modification of the pretrained visual stack are required.

Object, and Goal suites, which respectively stress spatial arrangement, object identity, and goal configuration. Each suite contains 10 language-conditioned tasks; we run 10 closed-loop episodes per task from benchmark-provided initial states, giving 100 episodes per suite and 300 per method. A separate Stage-3 policy is fine-tuned per method and per suite on that suite’s demonstrations. All variants and the baseline share the identical Stage-3 recipe and the identical evaluation configuration. We report task success rate (%). Full configurations are given in Appendices A and B.

Main results. Figure 3 and Table 3 present the main comparison. Both depth-observing variants improve over the baseline on all three suites. Model 2 attains the best average, 63.7% against the baseline’s 58.0%, and Model 4 follows at 62.7%.

Discrete tokens versus continuous features. The two targets do not order each other; they trade places. Model 2 leads on Spatial (+9) and Object (+5) but gains little on Goal (+3), while Model 4 shows the reverse profile: +9 on Goal and only +2 and +3 on Spatial and Object. A plausible reading follows the information each target retains. The

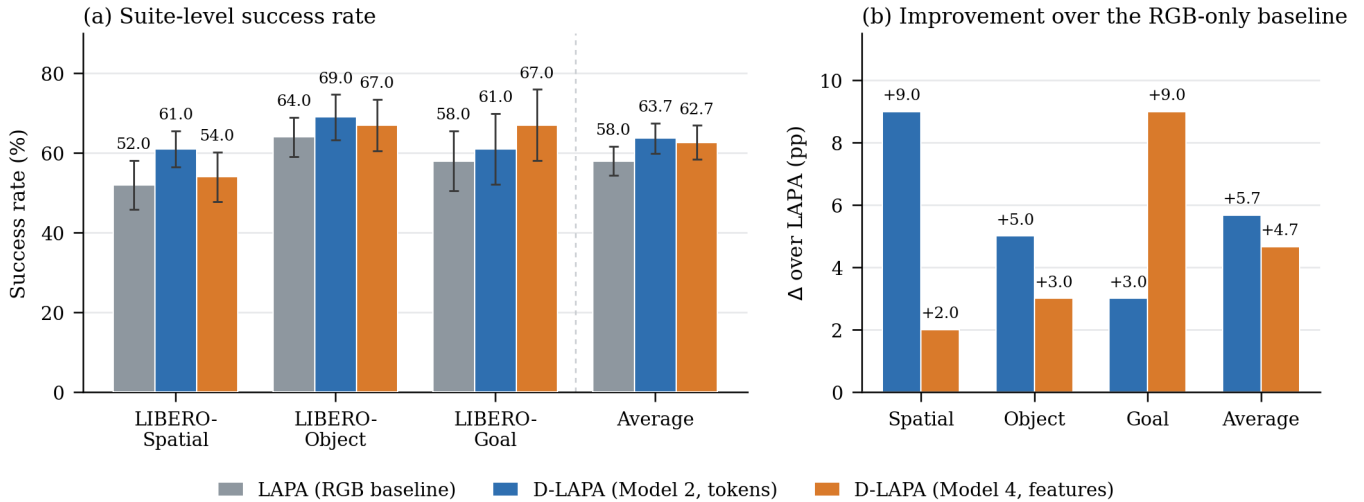


Figure 3: LIBERO results. **(a)** Suite-level success rate for the RGB-only LAPA baseline and the two depth-augmented variants, with the three-suite average at right. Error bars are standard errors of the mean across the 10 tasks in a suite (across all 30 tasks for the average), indicating the task-to-task spread underlying each score. **(b)** Absolute improvement over the baseline in percentage points. All six suite-by-variant cells are positive, but the two variants place their gains on different suites: token supervision on Spatial, feature distillation on Goal.

Table 3: Success rate (%) on LIBERO suites, each averaged over 10 tasks $\times$ 10 episodes. Best per column in **bold**. Model 5 results are pending; as the depth-free control, it is expected to land near the baseline.

Method	Spatial	Object	Goal	Avg.
LAPA baseline	52.0	64.0	58.0	58.0
D-LAPA (Model 2)	61.0	69.0	61.0	63.7
D-LAPA (Model 4)	54.0	67.0	67.0	62.7

Table 4: Absolute success-rate improvement, in percentage points, over the RGB-only LAPA baseline. Positive values denote improved performance. Suites are scored from 100 episodes each, so differences of a few points should not be read as separating the two variants.

Method	Spatial	Object	Goal	Average
Model 2	+9.0	+5.0	+3.0	+5.7
Model 4	+2.0	+3.0	+9.0	+4.7

token target compresses the depth transition into four codes from a size-8 codebook, producing a coarse, heavily regularized summary that discards magnitude and keeps categorical structure: which region moved, roughly where things sit relative to one another. LIBERO-Spatial, which holds objects and goals fixed and varies only their placement, is precisely a test of that categorical structure, and it is where the token variant gains most. The continuous target instead preserves the full 1024-dimensional pre-quantization feature, retaining graded distinctions the codebook throws away. LIBERO-Goal varies the commanded goal over related objects and configurations,

and several of its tasks turn on fine placement tolerances; this is where the continuous variant gains most. The reading is post hoc and rests on the two largest observed gaps, so we offer it as a hypothesis the design was not built to test, not as a result. What the results do support is the weaker and more useful statement: the choice of target appears to tune *where* the depth pathway helps, while both choices help on average.

Per-task structure of the gains. Suite means conceal a good deal, so the supplementary material breaks all 30 tasks out individually. The gains are not uniform lifts but redistributions. Counting per-task outcomes against the baseline, Model 2 wins on 16 tasks, ties on 7, and loses on 7; Model 4 wins on 13, ties on 8, and loses on 9. A positive suite mean therefore coexists with genuine per-task regressions. For example, LIBERO-Spatial task 6, which the baseline solves at 80%, drops to 50% under Model 2, and LIBERO-Goal task 8 falls from 30% to 20% under Model 4.

Feature analysis. Success rates say nothing about *how* the depth feature changes what the policy represents, so we probe the fine-tuned policies on LIBERO-Goal, the suite on which Model 4’s downstream gain is largest (+9 points): for each observation we extract the 4096-dimensional pre-action MLP feature and fit a ridge regression ($\lambda = 5000$) predicting the movement magnitude $|\Delta t|$ of the ground-truth action, repeat the probe on the policy’s predicted movement, and additionally measure how that magnitude is *arranged* using Moran’s I over the neighbourhood graph of the embedding. Two spaces can expose $|\Delta t|$ to a linear probe equally well while differing in whether neighbouring points share similar magnitudes. Both quantities are computed on the demonstrations the Stage-3 policy was trained on, so neither measures generalization; the probe on predicted movement is close to circular by construction, which is precisely what makes

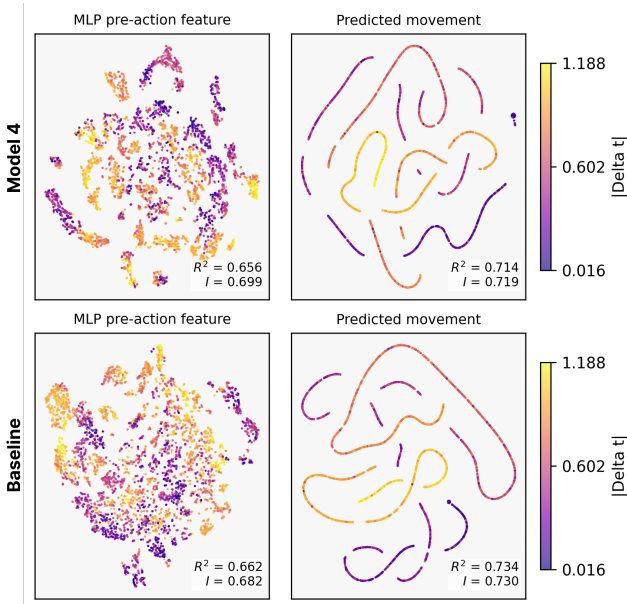


Figure 4: Feature analysis on LIBERO-Goal. Points are embedded observations, coloured by the ground-truth movement magnitude $|\Delta t|$ of the corresponding action. **Left:** the 4096-dimensional pre-action MLP feature. **Right:** the policy’s predicted movement. **Top:** D-LAPA (Model 4). **Bottom:** the RGB-only LAPA baseline. R^2 is the in-sample fit of a ridge probe ($\lambda = 5000$) from the representation to $|\Delta t|$; I is Moran’s I of $|\Delta t|$ over the embedding, measuring how locally coherent movement magnitude is. Both are computed on the demonstrations the policy was fine-tuned on.

it useful here, as it reads as fit to the demonstrated actions rather than as an independent quantity. Figure 4 locates the difference between the two policies not in the representation but in the mapping out of it. Upstream of the action head the two are near-identical in linear decodability ($R^2 = 0.656$ for D-LAPA against the baseline’s 0.662) and D-LAPA is the more locally coherent of the two ($I = 0.699$ against 0.682); downstream the ordering reverses, with the baseline fitting the demonstrated magnitudes more tightly on both statistics ($R^2 = 0.734$ versus 0.714; $I = 0.730$ versus 0.719). Since the action head is all that separates the two panels, the reversal is attributable to it: the baseline’s head gains +0.072 in R^2 and +0.048 in I across the final layer against Model 4’s +0.058 and +0.020, so it starts from a less coherent representation and arrives at a tighter fit, doing more of the work itself. A reasonable explanation is that a few demonstrations against a language-model backbone leaves memorization available to the action header, that the baseline exploits it, and that the depth feature which adds no new linearly decodable quantity but does make the representation vary more smoothly, leaves the head less to absorb, which would explain how tighter fit to the demonstrations coexists with worse behaviour in the environment and would suggest that under scarce robot data conditioning on depth alongside RGB is preferable to RGB alone.

Limitations. Beyond the pending control, four caveats apply. The evaluation covers a single benchmark family (Liu et al. 2023) and a single embodiment; transfer of the recipe to other simulators or real hardware is untested. Each suite is scored from 100 episodes with a single fine-tuning run per method and suite, so we cannot separate representation quality from seed-to-seed variation in Stage-3 optimization; repeated seeds are the most valuable cheap addition to this study. Because SSv2 carries no depth channel, both the teacher’s targets and the deployed input are monocular estimates rather than measurements, so the pathway supplies a geometric *prior* distilled from a depth estimator rather than new sensory information; whether a genuine depth sensor at either stage would widen the gain is untested and is a natural extension.

Conclusion

D-LAPA addresses a specific obstacle to geometry-aware vision-language-action models: not the absence of depth, but the absence of an encoder able to interpret it. Rather than presupposing a pretrained depth backbone, which unlike its RGB counterpart does not exist. We construct one by distillation. A depth latent-action teacher is trained once to model geometric change, a small single-frame student is distilled from its pseudo-labels, and the frozen policy receives one extra feature.

Two properties follow, and their conjunction is the claim we would most like carried forward. The pathway is learned entirely from unlabeled human demonstration video, with no robot trajectories, no action labels, and no depth sensor anywhere in the pipeline; and because the student reads a single frame, depth can be injected during online inference from monocular estimates of the policy’s own RGB observation. Individually these properties have precedents in recent depth-aware VLA work (Pang et al. 2024; Li et al. 2025a; Govind et al. 2026); together they remove both the data cost and the hardware cost that usually accompany geometric input, and make depth available to any frozen VLA with an ordinary camera.

Empirically, on LIBERO the recipe improves success in every suite-by-variant cell, lifting the three-suite average from 58.0% to 63.7%, with discrete and continuous supervision targets excelling on complementary task types. We are careful not to overclaim these gains, but the results suggest that a potential improvement in spatial understanding can be established solely from synthetic depth information of human demonstrations, with no robot data and no depth sensor, and apply that knowledge to build features that measurably improve downstream manipulation performance. Furthermore, we also introduce the architecture that provides a route by which depth reaches a pretrained VLA.

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